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Wake shaping apparatus and related technology

Abstract

The invention provides an apparatus that attaches to a hull of a boat to disrupt water flow around the hull such that a wake created behind the boat is larger, differently positioned, and/or differently shaped. In some embodiments, the apparatus is removably attached to the hull by one or more suction cups. The wake shaping apparatus can include a blade and be configured such that an angular orientation of the blade is adjustable. In certain embodiments, the wake shaping apparatus is inflatable.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. patent application Ser. No. 16/251,742, filed on Jan. 18, 2019, which is a continuation of U.S. patent application Ser. No. 15/858,779, filed on Dec. 29, 2017, and issued as U.S. Pat. No. 10,183,726, which claims priority to U.S. Provisional Patent Application No. 62/551,621, filed on Aug. 29, 2017, the entire contents of each of which are incorporated by reference herein.

FIELD OF THE INVENTION

(1) The present invention relates to a wake shaping apparatus. More specifically, the present invention provides a wake shaping apparatus configured to attach removably to a hull of a boat such that a wake created behind the boat is larger, differently positioned and/or differently shaped than when the boat is operated without the apparatus. The present invention also provides a deployable lanyard assembly that can be used with the wake shaping apparatus. Further, the invention provides a splash deflector.

BACKGROUND OF THE INVENTION

(2) When a boat moves through water, a wave pattern referred to as a wake is produced on the water surface. In some water sports, such as wake surfing, it is beneficial to modify the size, shape

and/or position of a wake created behind the boat in order to enhance the wake surfing experience.

(3) As set forth in the present disclosure, it would be desirable to provide a wake shaping apparatus that removably attaches to a hull of a boat. It would also be desirable to provide a wake shaping apparatus that overcomes the disadvantages of conventional wake shapers known in the prior art. The present invention overcomes such disadvantages, for example, by providing an apparatus that removably attaches to the hull of a boat, is easy to operate, is easily graspable, provides adjustability, and/or provides other advantages. It would also be desirable to provide a splash deflector that can be used with the wake shaping apparatus or on its own. Further, it would be desirable to provide a lanyard assembly that can be connected between the wake shaping apparatus and a boat.

SUMMARY OF THE INVENTION

(4) Certain embodiments of the invention provide an apparatus configured to attach removably to a hull of a boat. The apparatus comprises a base and a blade and has first and second configurations. The first configuration is characterized by the blade being mounted to the base. The second configuration is characterized by the blade being separated from the base. In the present embodiments, the apparatus is configured to provide tool-free adjustment between the first and second configurations.

(5) In some embodiments, the invention provides an apparatus having a handle configuration characterized by an aperture large enough to receive four fingers of a person's hand so as to facilitate manual handling of the apparatus. The apparatus is configured to attach removably to a hull of a boat.

(6) Some embodiments of the invention provide an apparatus having generally opposed first and second side regions and a lanyard eyelet. The lanyard eyelet is movable between at least first and second positions. When in the first position, the lanyard eyelet is located at the first side region of the apparatus. When in the second position, the lanyard eyelet is located at the second side region of the apparatus. The apparatus is configured to attach removably to a hull of a boat.

(7) Certain embodiments of the invention provide an apparatus having first and second configurations. The first configuration is characterized by the blade being mounted to the base. The second configuration is characterized by the blade being separated from the base. The blade has opposed first and second faces. When the apparatus is in the first configuration, the blade can be in a first orientation or a second orientation. In the first orientation, the first face of the blade faces a desired direction relative to the base. In the second orientation, the second face of the blade faces the desired direction relative to the base. The blade is equipped with a mount structure that projects outwardly from both of the first and second faces of the blade. The mount structure comprises a first detent that projects from the first face of the blade, and a second detent that projects from the second face of the blade. The first detent is configured to attach to the base of the apparatus such that the blade is retained in the first orientation. The second detent is configured to attach to the base of the apparatus such that the blade is retained in the second orientation. The apparatus is configured to attach removably to a hull of a boat.

(8) Further, some embodiments of the invention provide a combination of a boat and a splash deflector. The boat has a hull to which the splash deflector is mounted removably and noninvasively.

(9) Still further, certain embodiments provide an apparatus comprising an adjustable base. The adjustable base has first and second end regions. The first end region has a first suction cup, and the second end region has a second suction cup. The first end region has a pivot connection with the second end region. The apparatus is configured to attach removably to a hull of a boat.

(10) In other embodiments, the invention provides an apparatus comprising a base and a blade. The blade projects away from the base and has a plurality of channels. The apparatus is configured to attach removably to a hull of a boat.

(11) In still other embodiments, the invention provides an apparatus comprising a base and a blade.

The blade projects away from the base and has at least first and second orientations. When in the first orientation, the blade projects away from the base at a first angle. When in the second orientation, the blade projects away from the base at a second angle. The first and second angles are different. The apparatus is configured to attach removably to a hull of a boat.

(12) In certain other embodiments, the invention provides an apparatus configured to attach removably to a hull of a boat such that when attached to the hull, the apparatus is configured to disrupt water flow around the hull of the boat such that a wake created behind the boat during operation of the boat is larger, differently positioned, and/or differently shaped than when the boat is operated without the apparatus attached to the hull. The apparatus comprises a base, a blade and a lanyard assembly. The lanyard assembly comprises a tether portion and a float portion. The float portion comprises a float and a float line. The float line is configured to be attached to the base of the apparatus. The tether portion comprises a tether line configured to be attached to a cleat on the boat.

(13) In still other embodiments, the invention provides a combination of a boat and a wake shaping apparatus. The boat has a hull to which the wake shaping apparatus is mounted. The wake shaping apparatus is inflatable such that the wake shaping apparatus is configured to transition between an inflated state and a non-inflated state.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a broken-away perspective view of a wake shaping apparatus in accordance with certain embodiments of the present disclosure in which the apparatus is attached to a hull of a boat such that a front end of a base of the apparatus faces rearwardly of the boat.

(2) FIG. 2 is a broken-away perspective view of a wake shaping apparatus of the present disclosure similar to FIG. 1, except that the apparatus is attached to the hull of the boat such that a rear end of the base of the apparatus faces rearwardly of the boat.

(3) FIG. 3 is a side view of an embodiment of a wake shaping apparatus of the present disclosure attached to a hull of a boat such that a front end of a base of the apparatus faces rearwardly of the boat.

(4) FIG. 4 is a side view of a wake shaping apparatus of the present disclosure similar to FIG. 3, except that a rear end of a base of the apparatus faces rearwardly of the boat.

(5) FIG. 5 is a side view of an embodiment of a wake shaping apparatus of the present disclosure attached to a hull of a boat at a position that is nearer to a bow than to a stern of the boat.

(6) FIG. 6 is a side view of an embodiment of the present disclosure having two wake shaping apparatuses both attached to the same side of a hull of a boat such that one of the apparatuses is positioned near a bow of the boat and the other of the apparatuses is positioned near a stern of the boat.

(7) FIG. 7 is a front side perspective view of an embodiment of a wake shaping apparatus of the present disclosure in which the blade is mounted to the base.

(8) FIG. 8 is a top view of an embodiment of a wake shaping apparatus of the present disclosure.

(9) FIG. 9 is a front side perspective view of an embodiment of a wake shaping apparatus of the present disclosure in which the blade is separated from the base.

(10) FIG. 10 is a front view of an embodiment of a wake shaping apparatus of the present disclosure in which a body is in its second position.

(11) FIG. 11 is a front view of an embodiment of a wake shaping apparatus of the present disclosure in which the body is in its first position.

(12) FIG. 12 is a front perspective view of an embodiment of the blade of the present disclosure showing a convex face of the blade.

(13) FIG. 13 is a rear perspective view of an embodiment of the blade of the present disclosure showing a concave face of the blade.

(14) FIG. 14 is a rear view of an embodiment of a wake shaping apparatus of the present disclosure in which a lanyard eyelet is in the first position.

(15) FIG. 15 is a rear view of an embodiment of a wake shaping apparatus of the present disclosure in which the lanyard eyelet is in the second position.

(16) FIG. 16 is a bottom view of an embodiment of a wake shaping apparatus of the present disclosure showing two levers in a locked configuration.

(17) FIG. 17 is a partially exploded view of an embodiment of a wake shaping apparatus of the present disclosure showing individual components of two suction cup assemblies.

(18) FIG. 18 is a side view of an embodiment of a wake shaping apparatus of the present disclosure in which each of two levers is in an unlocked configuration.

(19) FIG. 19 is a top view of an embodiment of a wake shaping apparatus of the present disclosure.

(20) FIG. 20 is a top view of an embodiment of the present disclosure having an adjustable base that includes a pivot joint.

(21) FIG. 21 is side view of an embodiment of the present disclosure having an adjustable base that includes another pivot joint.

(22) FIG. 22 is a top view of an embodiment of the present disclosure having an adjustable base that includes a ball joint.

(23) FIG. 23 is a perspective view of a wake shaping apparatus and an embodiment of a splash deflector having an arcuate splash wall, both mounted removably to a hull of a boat.

(24) FIG. 24 is a front view of the splash deflector of FIG. 23.

(25) FIG. 25 is a bottom view of an embodiment of a splash deflector of the present disclosure.

(26) FIG. 26 is a perspective view of a wake shaping apparatus and an embodiment of a splash deflector having a planar splash wall, both mounted removably to a hull of a boat.

(27) FIG. 27 is a perspective view of an embodiment of a splash deflector of the present disclosure having an arcuate splash wall mounted to a transom of a boat.

(28) FIG. 28 is a side view of a blade, a partially exploded side view of an adapter component of the present disclosure, and a perspective view of a base showing a channel configured to selectively receive the first or second detent of the adapter component in certain embodiments of the present disclosure.

(29) FIG. 29 is a side view of an embodiment of the adapter component of the present disclosure showing the adapter component in each of first, second, and third positions.

(30) FIG. 30 is a perspective view of an embodiment of a wake shaping apparatus of the present disclosure in which a blade has an adjustable hinge.

(31) FIG. 31 is a perspective view of an embodiment of a wake shaping apparatus of the present disclosure in which a blade has a virtual hinge, as well as a side view showing the blade attached to the base.

(32) FIG. 32 is a top perspective view of an embodiment of a wake shaping apparatus of the present disclosure having a blade with channels.

(33) FIG. 33 is a detailed perspective view of the embodiment of FIG. 32.

(34) FIG. 34 is a perspective view of an embodiment of both a wake shaping apparatus and a deployable lanyard assembly in which a float portion of the lanyard assembly is attached to the wake shaping apparatus and a tether portion of the lanyard assembly attached to a cleat on a boat.

(35) FIG. 35 is a perspective view of the lanyard assembly of FIG. 34 where the float portion and the tether portion are separated.

(36) FIG. 36 is a side detail view of the deployable lanyard assembly of FIG. 34.

(37) FIG. 37 is a partially exploded side view of the deployable lanyard assembly of FIG. 36.

(38) FIG. 38 shows two views of an embodiment of an inflatable wake shaping apparatus in inflated and non-inflated states so as to schematically illustrate how the apparatus transitions

between the two states.

(39) FIG. **39** shows multiple views of another embodiment of an inflatable wake shaping apparatus in both inflated and non-inflated states so as to schematically illustrate how the apparatus transitions between the two states.

(40) FIG. **40** shows multiple views of one or more inflatable wake shaping apparatuses attached to different regions of a boat.

(41) FIG. **41** is a schematic perspective view of an embodiment of a wake shaping apparatus of the present disclosure in which the wake shaping apparatus is an inflatable apparatus.

(42) FIG. **42** is a schematic rear view of the wake shaping apparatus of FIG. **41**.

(43) FIG. **43** is a schematic side view of another embodiment of a wake shaping apparatus of the present disclosure in which the wake shaping apparatus is an inflatable apparatus.

(44) FIG. **44A** is a schematic side view of still another embodiment of a wake shaping apparatus of the present disclosure in which the wake shaping apparatus is an inflatable apparatus.

(45) FIG. **44B** is a schematic side view of yet another embodiment of a wake shaping apparatus of the present disclosure in which the wake shaping apparatus is an inflatable apparatus.

(46) FIG. **44C** is a schematic side view of still another embodiment of a wake shaping apparatus of the present disclosure in which the wake shaping apparatus is an inflatable apparatus.

(47) FIG. **45** is a broken-way rear view of a wake shaping apparatus attached to an underside of a hull of a boat in accordance with certain embodiments of the present disclosure.

DESCRIPTION OF VARIOUS EMBODIMENTS

(48) The following detailed description is to be read with reference to the drawings, in which like elements in different drawings have like reference numerals. The drawings, which are not necessarily to scale, depict selected embodiments and are not intended to limit the scope of the invention. Skilled artisans will recognize that the examples provided herein have many useful alternatives that fall within the scope of the invention.

(49) Referring to the drawings, and in particular, FIG. **1**, there is shown an apparatus of the present disclosure generally represented by reference numeral **10**. The illustrated apparatus **10** is a wake shaping apparatus, which preferably is configured to attach removably to a hull **40** of a boat **50**. The wake shaping apparatus **10** is configured to disrupt water flow around the hull **40** such that a wake created behind the boat **50** during operation of the boat **50** is larger, differently positioned, and/or differently shaped than when the boat **50** is operated without the apparatus **10** attached to the hull **40**.

(50) As shown in FIG. **7**, the apparatus **10** generally comprises a base **100** and a blade **300**. The apparatus **10** has a first configuration **20** in which the blade **300** is mounted to the base **100**, and a second configuration **30** in which the blade **300** is separated from the base **100**. As described in greater detail below, in some embodiments, the apparatus **10** is configured to provide tool-free adjustment between the first **20** and second **30** configurations.

(51) The base **100** generally has first **115** and second **120** end regions, and first **125** and second **130** side regions. The illustrated first **125** and second **130** side regions extend between the first **115** and second **120** end regions. The first end region **115** includes a front end **135** (see FIG. **3**) of the base **100**, and the second end region **120** includes a rear end **140** of the base **100**.

(52) The base **100** also has a top side **145** and a bottom side **148** (see FIG. **16**). The top side **145** is opposite the bottom side **148**. Preferably, the bottom side **148** is on the same side of the base **100** as one or more suction cups **105**. When the apparatus **10** is attached to the hull **40** of a boat **50**, the bottom side **148** faces (and is carried alongside) the hull **40**, and the top side **145** faces away from the hull **40**.

(53) Referring to FIG. **8**, the blade **300** has opposed first **305** and second **310** faces. In some embodiments, the first **305** and second faces **310** are identical, whereas in other embodiments, they are different. For example, the first face **305** of the blade **300** can have different structural features provided thereon and/or a different shape than the second face **310** of the blade **300**. Such

differences can optionally be provided to create different effects on the wake. In certain embodiments, one of the faces **305, 310** is convex (FIG. 12) and the other of the faces **305, 310** is concave (FIG. 13). However, in other embodiments, both faces **305, 310** of the blade **300** are planar. Alternatively, one face **305, 310** can be planar while the other face **305, 310** is either concave or convex.

(54) As discussed above, the illustrated apparatus **10** has a first configuration **20** (see FIG. 7) in which the blade **300** is mounted to the base **100**. When the apparatus **10** is in the first configuration **20**, the blade **300** is in either a first orientation or a second orientation. The first face **305** of the blade **300** faces a desired direction relative to the base **300** when the apparatus **10** is in the first orientation. In contrast, the second face **310** of the blade **300** faces the desired direction relative to the base when the apparatus **10** is in the second orientation. It may be advantageous to change the blade **300** between the first orientation and the second orientation when the first **305** and second **310** faces of the blade **300** have different structures and/or shapes in order to achieve different effects on the wake.

(55) The illustrated blade **300** has a mount structure **340** configured to attach to the base **100**. The mount structure **340** preferably projects outwardly from the blade **300** (e.g., outwardly from a primary wall portion of the blade). For example, the mount structure **340** can optionally project outwardly from at least one, and preferably both, of the first **305** and second **310** faces of the blade **300**. In some alternate embodiments, the base **100** is integral to (or permanently affixed to) the blade **300**. However, in preferred embodiments, the base **100** and the blade **300** are configured to be attached to each other removably. Where the mount structure **340** projects outwardly from both faces **305, 310** of the blade **300**, the blade **300** is configured to be removably attached to the base **100** in either of two orientations, i.e., in either the first or second orientation so that a desired one of the faces **305, 310** of the blade **300** faces the desired direction.

(56) Preferably, the apparatus **10** is configured to be adhered to the hull **40** of the boat **50** by at least one suction cup **105** so as to removably attach the apparatus **10** to the hull **40** of the boat **50**. However, the apparatus **10** can alternatively be removably attached to the hull **40** through other non-invasive forms of mechanical attachment, such as hook and loop fasteners, clasps, magnets, or the like. The removable nature of the apparatus **10** allows it to be used on a boat **50** that already has a built-in wake-shaping device so as to complement the effects provided by the built-in wake-shaping device. Alternatively, the apparatus **10** can be used on a boat **50** that does not have any other wake-shaping devices.

(57) Thus, the apparatus **10** preferably can be attached removably to the boat **50** in a desired orientation and on a desired region of the boat **50**. Both the orientation of the apparatus **10** when attached to the boat, and the region of the boat to which the apparatus **10** is attached, can be selected to achieve particular effects on the wake. For example, in FIGS. 1 and 3, the apparatus **10** is attached to the hull **40** of the boat **50** such that the front end **135** of the base **100** faces rearwardly of the boat **50**. FIGS. 2 and 4 show the reverse orientation, i.e., the rear end **140** of the base **100** faces rearwardly of the boat **50**. The apparatus **10** can be attached to the hull **40** of the boat **50** at a position that is adjacent to a stern **90** (FIGS. 1-4) or adjacent to a bow **95** (FIG. 5) of the boat **50**. Additionally or alternatively, a wake shaping apparatus **10** can be attached to a bottom (or underside) **40U** of the hull **40**. Reference is made to FIG. 45. Any embodiment of a wake shaping apparatus **10** described in this disclosure can optionally be attached releasably to the bottom **40U** of the hull **40**. This can optionally be in addition to there being one or more wake shaping apparatuses provided on other locations of the hull. Alternatively, the only wake shaping apparatus **10** attached releasably to the hull may be on the bottom (or underside) **40U** of the hull **40**. This can optionally be the case in FIG. 45. While FIG. 45 shows the wake shaping apparatus **10** as being an inflatable apparatus **10b**, this is by no means required. To the contrary, FIG. 45 represents a group of embodiments wherein any wake shaping apparatus is attached releasably to the bottom (or underside) of a hull of a boat. In FIG. 45, the illustrated wake shaping apparatus **10** is attached to

the bottom of the hull by a plurality of suction cups. However, various other releasable attachment means can be used instead.

(58) In some embodiments, two wake shaping apparatuses **10** are each attached to the same side of a hull **40** of a boat **50**. As shown in FIG. **6**, one of the apparatuses **10** can be positioned adjacent to the bow **95** of the boat **50**, while the other apparatus **10** is positioned adjacent to the stern **90** of the boat **50**. Given the present teaching as a guide, skilled artisans will appreciate that the one or more apparatuses **10** can be mounted removably at various locations on the hull.

(59) Tool-Free Adjustment

(60) In certain embodiments, the base **100** and the blade **300** can be separated from each other to achieve the second configuration **30** described above. One example of the second configuration **30** is shown in FIG. **9**. Preferably, when the apparatus **10** is in the second configuration **30**, the base **100** and the blade **300** can be nested alongside each other so as to occupy a more compact (i.e., smaller) space than when the apparatus **10** is in the first configuration **40**. In certain embodiments, the apparatus **10** when in the second configuration can have its blade and base nested alongside each other so as to fit within a rectangular volume (e.g., inside a box) having the following dimensions: 15 inches by 6 inches by 9 inches, or even 14 inches by 5 inches by 8 inches. While not in use, the apparatus **20** can be removed from the hull **40** of the boat **50** and stored conveniently until needed.

(61) In some embodiments, the apparatus **10** can be adjusted from the first configuration **20** to the second configuration **30**, and from the second configuration **30** to the first configuration **20**, without using any tools, and preferably without removing or inserting any fastener from or into the base **100**. In such cases, the blade **300** can be mounted to the base **100** and later removed therefrom without using any tools. This is particularly advantageous since the apparatus **10** is intended to be attached to the hull of a boat **50** at a location that is under water while the boat **50** is in operation. Using tools underwater can prove difficult due to visibility issues, the tendency of tools to become slippery when wet, and/or the potential for the water to be cold. These factors may lead to tools and/or fasteners being dropped into the water.

(62) In some embodiments, an optional hook and loop fastener is sandwiched between the blade **300** and the base **100** when the apparatus **10** is in the first configuration **20**. In such embodiments, the hook and loop fastener is configured to removably fasten the blade **300** to the base **100**. When provided, the hook and loop fastener preferably is used in addition to the mount structure **340**. However, in some embodiments, the hook and loop fastener may replace the mount structure **340** entirely.

(63) When provided, the hook and loop fastener includes a first portion attached to the base **100** and a second portion attached to either the mount structure **340** or directly to one or both faces **305**, **310** of the blade **300**. Preferably, the second portion of the hook and loop fastener is attached to the mount structure **340** on both the first **305** and second **310** faces of the blade **300** so that the blade **300** can be positioned in either the first **315** or second **320** orientation as desired.

(64) As discussed above, it is preferable that the apparatus **10** is configured to be removably attached to a hull of a boat using one or more suction cups **105**. In such cases, at least one suction cup **105** on the base **100** forms part of a suction cup assembly **110**. Preferably, the suction cup assembly **110** includes at least two (or has exactly two) suction cups **105**. However, the suction cup assembly **110** can alternatively include only one suction cup **105**, or it can include more than two suction cups **105**.

(65) In preferred embodiments, the base **100** further includes a detent **150** having at least first **155** and second **160** positions. In the first position **155**, the detent **150** is configured to lock the blade **300** on the base **100**. In the second position **160**, the detent **150** is configured to release the blade **300** from the base **100**. One example of the second position **160** is shown in FIG. **10**, and one example of the first position **155** is shown in FIG. **11**. Preferably, the detent **150** is rotatable between its first **155** and second **160** positions. However, it should be appreciated that in some

embodiments, the detent **150** is not limited to rotational movement, and can instead move linearly, for example. In some embodiments, the detent **150** is part of a suction cup assembly **110** and is located between the base **100** and a suction cup **105**.

(66) The illustrated base **100** (e.g., a shoulder thereof) has a channel **165** therein. In a preferred embodiment, the channel **165** is a slot. The illustrated blade **300**, and more specifically an adapter component **325** of the illustrated blade, comprises at least one plate, and preferably two plates **355**, **365**, each configured to be received removably in the channel **165**. The adapter component **325** and the channel **165** are shown in FIG. **28**. Preferably, the two plates of the adapter component **325** include a first plate **355** disposed adjacent the first face **305** of the blade **300**, and a second plate **365** disposed adjacent the second face **310** of the blade **300**. The adapter component **325**, however, can alternatively have only a single plate, which can be located on either the first face **305** or second face **310** of the blade **300**.

(67) The base **100**, and more specifically the detent **150**, can comprise a body **170** that is moveable between first **175** and second **180** positions. When in the first position **175** (see FIG. **11**), the body **170** at least partially closes the channel **165** of the base **100**. When in the second position **180** (see FIG. **10**), the body **170** is spaced from the channel **165** of the base **100**. In a preferred embodiment, the body **170** comprises a rotatable flange **185** that is rotatable between its first **175** and second **180** positions. However, it should be appreciated that in some embodiments, the body **170** is not limited to rotational movement, and can instead move linearly, for example. The body **170** can optionally comprise a generally circular disc **190** from which the rotatable flange **185** projects. In preferred embodiments, the body **170** is part of a suction cup assembly **110** and the disc **190** is located between a suction cup **105** and the base **100**.

(68) When provided, the suction cup assembly **110** can optionally include at least one lever **195** moveable between first **200** and second **205** positions. Preferably, one lever **195** is coupled to each respective suction cup **105** so as to be operable to activate and release suction for the respective suction cup **105**. However, it is possible that a single lever could be provided that simultaneously controls the suction force for all or a select group of multiple suction cups. Also, as noted above, there may be only a single suction cup on the apparatus in some cases. When the one or more levers **195** are each in the first position **200** (see, e.g., FIG. **8**), each suction cup **105** is in a locked configuration. The locked configuration helps secure the respective suction cup **105** to the hull **40** of the boat **50**. When the at least one lever **195** is in the second position **205** (see FIG. **18**), the respective suction cup **105** is in an unlocked configuration. The unlocked configuration allows the respective suction cup **105** to be released from the hull **40** of the boat **50**. Two levers **195** are each shown in a locked configuration in FIGS. **7**, **8** and **19**, and in an unlocked configuration in FIG. **18**.

(69) In some embodiments, two suction cups **105** are disposed in a confronting angle configuration. In such cases, the confronting angle configuration is characterized by a tilt angle (e.g., relative to horizontal and/or desired plane) of greater than zero degrees and up to three degrees. By having two suction cups **105** in such a confronting angle configuration, the suction cups **105** may adhere particularly stably to a curved surface of a hull **40** of a boat **50**.

(70) It is to be appreciated that tool-free adjustment need not be provided in all embodiments of the invention. However, any embodiment of the present disclosure can optionally be provided with tool-free adjustment of the nature described above.

(71) Handle Configuration

(72) In some embodiments, the apparatus **10** has a handle configuration **210** characterized by an aperture **215** in the apparatus **10**. Preferably, the aperture **215** is in the base **100**, as shown in FIG. **16**. The aperture **215** is sized to facilitate manual handling and grasping of the apparatus **10**. In a preferred embodiment, the aperture **215** is large enough (e.g., is configured) to receive four fingers of a person's hand. Preferably, the aperture **215** is elongated and/or has a generally rectangular or generally oval shape. However, it is contemplated that the aperture **215** can have any shape as long as it does not interfere with the aperture's intended purpose. In certain embodiments, the aperture

has a length of at least two inches, or at least three inches, such as between three inches and six inches.

(73) In a preferred embodiment, the aperture **215** has an aspect ratio of greater than two. As used in the present disclosure, the aspect ratio is defined as a length of the aperture **215** divided by a width of the aperture **215**. The length of the aperture **215** is measured along an axis extending between the front end **135** and the rear end **140** of the base **100**. The width of the aperture **215** is measured along an axis extending between the first **115** and second **120** side regions of the base **100**.

(74) The base **100** preferably has two ribs **230** that extend between (optionally entirely between) the front end **135** and rear end **140** of the base **100**. In preferred embodiments, the aperture **215** is located (e.g., defined) between the ribs **230**. Preferably, both of the ribs **230** are defined by a single integral body. However, in other embodiments, the ribs **230** can be separate and distinct structures from each other. Each rib **230** preferably is elongated, e.g., so as to have a length longer than six inches, such as between six and 14 inches.

(75) As discussed above, the base **100** has a top side **145** and a bottom side **148**. Preferably, the aperture **215** passes entirely through the base **100** from its bottom side **148** to its top side **145**. However, in other embodiments, the aperture **215** is a blind opening such that the aperture **215** passes only partially between the bottom side **148** and the top side **145** of the base **100** (e.g., so as to have a closed bottom).

(76) As also discussed above, in some embodiments, the suction cup assembly **110** includes two levers **195**, each of which is associated with a respective suction cup **105**. Such levers **195** can advantageously be disposed in a recessed area **220** (see FIG. 7) of the base **100**. The recessed area **220** can, for example, define a channel **225** that extends between two ribs **230** of the base **100** and is open to aperture **215**. Both of the illustrated levers **195** are aligned with the aperture **215** when in their first (locked) positions **200**. Preferably, the levers **195** do not hinder (or at least do not prevent) manual access to the aperture **215**. Additionally, when in the first position **200**, the levers **195** can optionally be located entirely within the recessed area **220**, e.g., so that the levers **195** are sheltered from being inadvertently moved to the second position **205** and thus unlocked.

(77) It is to be appreciated that the handle configuration need not be provided in all embodiments of the invention. However, any embodiment of the present disclosure can optionally be so configured.

(78) Lanyard Eyelet

(79) The apparatus **10** preferably has a lanyard eyelet **235**. In such cases, in order to tie the apparatus **10** to a boat **50**, a tether **500** can be removably attached to both the boat **50** and the lanyard eyelet **235**. The tether **500** can comprise a rope, bungee cord or the like.

(80) When provided, the lanyard eyelet **235** preferably is movable between at least first **240** and second **245** positions. When in the first position **240**, the lanyard eyelet **235** is located at the first side region **125** of the apparatus **10**. When in the second position **245**, the lanyard eyelet **235** is located at the second side region **130** of the apparatus **10**. FIG. 14 shows the lanyard eyelet **235** in the first position **240**, and FIG. 15 shows the lanyard eyelet **235** in the second position **245**.

Providing for such movement of the lanyard eyelet **235** is advantageous since it can be positioned as needed for convenient attachment thereto by a tether **500**. For example, when the front end **135** of the base **100** faces rearwardly of the boat **50**, the lanyard eyelet **235** can be moved to the first position **240**, and thus, to the first side region **125** of the base **100**. On the other hand, when the rear end **140** of the base **100** faces rearwardly of the boat **50**, the lanyard eyelet **235** can be moved to the second position **245**, and thus, to the second side region **130** of the base **100**. This may make it possible to use a shorter length of tether, which may be advantageous. It can also help keep the tether up out of the water flow path along the wake shaping apparatus.

(81) Preferably, the apparatus **10** includes a suction cup assembly **110** comprising a disc **250** that is rotatable between first **240** and second **245** positions (see FIGS. 14 and 15). In preferred embodiments of this nature, the lanyard eyelet **235** can optionally be part of the suction cup

assembly **110** and may be located between a suction cup **105** and the base **100**. In such cases, the lanyard eyelet **235** preferably is a projection of the rotatable disc **250**.

(82) In another embodiment, the suction cup assembly **10** does not include rotatable disc **250**. In this embodiment, the lanyard eyelet **235** can be fixedly or removably attached to any portion of the apparatus **10**.

(83) In embodiments where the apparatus **10** includes a suction cup assembly **110** having two suction cups **105**, the rotatable disc **250** and a generally circular disc **190** can each be positioned in a stacked orientation and located between a respective suction cup **105** and the base **100**.

Alternatively, in another embodiment having only a single suction cup **105**, rotatable disc **250** can be omitted, while a generally circular disc **190** is positioned between the single suction cup **105** and the base **100**.

(84) A detailed view of one example of a suction cup assembly **110** is shown in FIG. **17**. Here, each lever **195** of the suction cup assembly **110** has a hole **505** disposed therein. A pin **510** is provided for each lever **195** such that one of the pins **510** can be received in the hole **505** of the respective lever **195**. Each suction cup **105** further has a post **515** attached thereto and extending outwardly therefrom. Each post **515** has a hole **520** disposed therein. In addition to passing through the hole **505** of the respective lever **195**, each pin **510** also passes through the hole **520** of the post **515** of the respective suction cup **105**. Additional holes (not shown) are provided in the base **100**, disc **190**, and disc **250** to allow the levers **195** to attach to the respective suction cups **105**. The suction cup assembly **110** also includes springs **525**, optionally such that one spring **525** is provided for each lever **195**. The springs **525** are located between the levers **195** and their respective suction cups **105**.

(85) It is to be appreciated that the lanyard eyelet need not be provided in all embodiments of the invention. However, any embodiment of the present disclosure can optionally be provided with an adjustable lanyard eyelet of the nature described above.

(86) Mount Structure

(87) As discussed above, the apparatus **10** preferably has different configurations depending on whether the base **100** is mounted to the blade **300** (first configuration **20**) or separated from the blade **300** (second configuration **30**). Additionally, when the apparatus **10** is in the first configuration **20**, the blade **300** preferably is in either a first orientation or a second orientation, depending on which face **305**, **310** of the blade **300** faces the desired direction relative to the base **100**.

(88) As also discussed above, the blade **300** can advantageously be equipped with a mount structure **340** that projects outwardly from at least one, and preferably both, faces **305**, **310** of the blade **300**. In some embodiments, the mount structure **340** comprises a first detent **345** that projects from the first face **305** of the blade **300**, and a second detent **350** that projects from the second face **310** of the blade **300**. The first detent **345** is configured to attach to the base **100** of the apparatus **10** such that the blade **300** is retained in the first orientation **315**. The second detent **350** is configured to attach to the base **100** of the apparatus **10** such that the blade **300** is retained in the second orientation **320**.

(89) In certain embodiments, the first **345** and second **350** detents are fixed in place on (i.e., so as not to be movable relative to) the blade **300**. In other embodiments, however, the first **345** and second **350** detents are part of an adapter component **325** that is moveable (e.g., relative to a primary wall of the blade **300**) between at least first **330** second **335** positions. In the first position **330**, the adapter component **325** is configured to attach to the base **100** of the apparatus **10** such that the blade **300** is retained in the first orientation **315**. In the second position **335**, the adapter component **325** is configured to attach to the base **100** of the apparatus **10** such that the blade **300** is retained in the second orientation **320**.

(90) In some embodiments, the blade **300** is configured to lie at least generally in a plane (e.g., has a configuration that is at least generally planar), and the adapter component **325** is configured to

move at least generally perpendicular to such plane when the adapter component **325** moves between its first **330** and second **335** positions. The adapter component **325** can also move to an intermediate position **338** that is in-between the first **330** and second **335** positions. The first **345** and second **350** detents can optionally be connected together by conventional fasteners, such as nuts and bolts. Such fasteners can allow the first **345** and second **350** detents to slide between the first **330** and second **335** positions.

(91) As shown in FIG. **29**, when the adapter component **325** is in the first position **330**, the first detent **345** projects further from the first face **305** of the blade **300** than it does when the adapter component **325** is in the second position **335**. Likewise, when the adapter component **325** is in the second position **335**, the second detent **350** projects further from the second face **310** of the blade **300** than it does when the adapter component **325** is in the first position **330**. FIG. **29** shows the adapter component **325** in the first position **330** (image on the left), the second position **335** (middle image), and the intermediate position **338** (image on the right).

(92) The first detent **345** of the adapter component **325** can optionally comprise a first plate **355** having a first mounting flange **360**. Similarly, the second detent **350** of the adapter component **325** can optionally comprise a second plate **365** having a second mounting flange **370**. Preferably, the first **355** and second **365** plates are moveable conjointly relative to the first **305** and second **310** faces of the blade **300**. When the blade **300** is positioned in the first orientation **315**, the first mounting flange **360** is received in the channel **165**. When the blade **300** is positioned in the second orientation **320**, the second mounting flange **370** is received in the channel **165**.

(93) As discussed above, the base **100** has opposed first **115** and second **120** end regions. The first end region **115** (e.g., a shoulder **255** thereof) defines the channel **165**. In a preferred embodiment, the first end region **115** has a greater height than the second end region **120**. Alternatively, in some embodiments, the first and second end regions can be identical in height, or the second end region can have a taller height than the first end region so long as the second end region includes the channel.

(94) It is to be appreciated that the preferred blade mount structure need not be provided in all embodiments of the invention. However, any embodiment of the present disclosure can optionally be provided with a blade mount structure of the nature described above.

(95) Adjustable Base

(96) In certain embodiments, the base **100** is an adjustable base **100a**. In some embodiments of this nature, the first end region **115** of the adjustable base **100a** has a first suction cup **105a**, and the second end region **120** of the adjustable base **100a** has a second suction cup **105b**. To make such a base adjustable, the first end region **115** can optionally have a pivot connection **260** with the second end region **120**. In some cases, the pivot connection **260** allows the suction cups **105a**, **105b** to be oriented at various angles relative to each other. In this manner, the pivot connection **260** may permit the suction cups **105a**, **105b** to be stably mounted to surfaces of different contours, including both flat and curved surfaces. For example, the suction cups **105a**, **105b** may be configured to be positioned in a confronting angle configuration such that the suction cups **105a**, **105b** are angled somewhat inwardly toward each other.

(97) When provided, the pivot connection **260** preferably comprises a first pivot joint having a first pivot axis. In FIG. **20**, the first pivot joint enables adjusting an angular orientation of the second end region **120** relative to that of the first end region **115** such that the apparatus **10** has at least first and second angular configurations. The first angular configuration is characterized by the first and second end regions extending along a straight line (see FIG. **19**). The second angular configuration is characterized by one of the first **115** and second **120** end regions projecting away from the other at a positive acute angle (see FIG. **20**). In this embodiment, both the first **105a** and second **105b** suction cups can remain in a constant plane while the apparatus **10** is moved between its first and second angular configurations.

(98) Referring now to FIG. **21**, the pivot connection **260** can optionally comprise a second pivot

joint having a second pivot axis. In other embodiments, the pivot connection **260** includes both the first and second pivot joints. In FIG. **21**, the pivot connection has only the second pivot joint. The second pivot joint enables pivoting the apparatus **10** in either a first direction or a second direction. In particular, when the apparatus **10** is pivoted in the first direction, the first **105a** and second **105b** suction cups pivot toward each other. When the apparatus **10** is pivoted in the second direction, the first **105a** and second **105b** suction cups pivot away from each other.

(99) The pivot connection **260** can optionally comprise a ball joint, as shown in FIG. **22**. When provided, the ball joint preferably is located between the first **115** and second **120** end regions of the adjustable base **100a** and preferably is configured to enable pivotal movement between the first **115** and second **120** end regions.

(100) It is to be appreciated that the adjustable base structure need not be provided in all embodiments of the invention. However, any embodiment of the present disclosure can optionally be provided with an adjustable base structure of the nature described above.

(101) Splash Deflector

(102) The invention also provides embodiments wherein a splash deflector **265** is mounted removably and non-invasively to a boat **50**. The splash deflector **265** is intended to limit the splash of water adjacent to a region of the boat **50** where the splash deflector **265** is attached. The illustrated boat **50** has a transom **60** that extends between opposed sides **70**, **80** of the hull **40**. The splash deflector **265** can be mounted to the transom **60**, as shown in FIG. **27**, or to a desired one of the sides **70**, **80** of the hull **40**, as shown in FIGS. **23** and **26**.

(103) If desired, first and second splash deflectors can be mounted respectively on first and second sides of the hull. Additionally or alternatively, first and second splash deflectors can be mounted on the transom of a boat. Given the present teaching as a guide, skilled artisans will appreciate that one or more splash deflectors can be provided at various locations on one or both sides of the hull and/or at various locations on the transom.

(104) The splash deflector **265** comprises both a splash wall **270** and a mount wall **275**. The splash wall **270** is attached to and projects outwardly from the mount wall **275**. When the splash deflector **265** is attached to the hull **40** of a boat **50**, the mount wall **275** is carried alongside the hull **40**, and the splash wall **270** projects away from the hull **40**. As shown in FIG. **26**, the splash wall **270** can be planar, or at least generally planar, such that it extends away from the mount wall **275** in a generally perpendicular manner. Alternatively, as shown in FIGS. **23** and **27**, the splash wall **270** can be arcuate, e.g., so as to form a curved fender configuration.

(105) The splash deflector **265** is removably mounted to a boat **50**, preferably by one or more suction cups **280**. In a preferred embodiment, the mount wall **275** of the splash deflector **265** includes two suction cups **280**, as shown in FIG. **25**. However, in other embodiments, the splash deflector **280** has only a single suction cup **280**, or it has more than two suction cups **280**. Moreover, instead of suction cups **280**, the splash deflector **265** can be attached to the boat by other non-invasive and removable forms of mechanical attachment, such as hook and loop fasteners, magnets or the like.

(106) Similar to the wake shaping apparatus **10**, the splash deflector **265** can optionally include one or more levers **285** that control suction of the preferred suction cups **280**. Preferably, one lever **285** is provided for each suction cup **280**, as shown in FIG. **24**. However, a single lever could alternatively be provided that controls suction for all suction cups on the mount wall. The lever **285** can be moved between a locked configuration and an unlocked configuration. In the locked configuration, each lever **285** activates suction for the respective suction cup **280**. In the unlocked configuration, each lever **285** releases suction for the respective suction cup **280**.

(107) Preferably, the splash deflector **265** has an aperture **290** passing therethrough. When provided, the aperture **290** preferably is sized to facilitate manual handling and grasping of the splash deflector **265**. In such cases, the aperture **290** is configured to receive four fingers of a person's hand. The mount wall **275** has a first surface **295** and a second surface **298**. Preferably, the

aperture **290** passes entirely through the mount wall **275** from its first surface **295** to its second surface **298**, as shown in FIG. **23**. In other embodiments, however, the aperture comprises a blind bore such that the aperture passes only partially between the first surface and the second surface of the mount wall

(108) The splash deflector **265** can be used with or without the wake shaping apparatus **10**. When the splash deflector **265** and the wake shaping apparatus **10** are used in combination, the splash deflector **265** is positioned above the apparatus **10**, as shown in FIG. **27**. Such positioning of the splash deflector **265** can advantageously limit the amount of water from the apparatus **10** that splashes into the boat **50**.

(109) In certain embodiments, the splash deflector and the wake shaping apparatus are two discrete components that can be removably attached to each other during use. Removable attachment of the splash deflector and the wake shaping apparatus can be accomplished by various conventional means, including, but not limited to, a mounting wall (not shown) attached to both the splash deflector and the apparatus by screws or other fasteners.

(110) In other embodiments, the splash deflector and the apparatus are integrally formed such that they define a one-piece structure. For example, a mounting wall can be provided that fixedly attaches the splash deflector to the wake shaping apparatus.

(111) Adjustable Blade

(112) In some embodiments, an angular orientation of the blade **300** (e.g., relative to the base **100**) can be adjusted. This feature allows the blade **300** to be adjusted so as to project away from the base **100** at different angles to provide various effects on the wake. In the present embodiments, the blade **300** has at least first and second orientations. In the first orientation, the blade **300** projects away from the base **100** at a first angle. In the second orientation, the blade **300** projects away from the base **300** at a second angle. The first and second angles are different. Preferably, the blade **300** is pivotable between the first and second orientations via a hinge **380**. The hinge **380** can be a simple hinge, as shown in FIG. **30**, or a virtual hinge, as shown in FIG. **31**.

(113) The blade **300** further includes a mounting plate **385**. The mounting plate **385** is attached to the blade **300** (e.g., is attached to a primary wall portion of the blade), e.g., via the hinge **380**, preferably such that the hinge **380** enables the blade **300** (or a primary wall portion thereof) to pivot relative to the mounting plate **385**. Similar to the first **355** and second **360** plates described above, the mounting plate **385** can be removably mounted in a channel **165** defined by the base **100**. In some embodiments, the mounting plate **385** is attached directly to the blade **300** (e.g., to a primary wall portion thereof) via the hinge **380**.

(114) As shown in FIG. **30**, an optional base plate **390** can be provided. The base plate **390**, for example, can be attached to the blade **300** (e.g., to a primary wall portion thereof) between the blade **300** (e.g., a primary wall portion thereof) and the mounting plate **385**. In FIG. **30**, the mounting plate **385** is hingedly attached to the base plate **390** such that the mounting plate **385** is pivotable relative to the base plate **390**.

(115) As discussed above, FIG. **31** shows a virtual hinge allowing the blade **300** to pivot between the at least first and second orientations. In some embodiments having a virtual hinge, the mounting plate **385** has a projection **395** extending outwardly therefrom. The projection **395** of the mounting plate **385** can have a slot **398** disposed therein. Preferably, the projection **395** is a semi-circle, and/or the slot **398** is arcuate, as shown in FIG. **31**. However, other shapes for both the projection **395** and the slot **398** are contemplated.

(116) In a preferred embodiment, a knob and bolt structure is provided. In certain embodiments, two walls (not shown) are attached to and extend between the blade **300** and the mounting plate **385**. One or more bolts can be attached to the walls. One or two knobs can be provided, and each knob is attached to an end of the respective bolt. The bolt(s) are received in the slot **398** and are movable along a length of the slot **398**. Tightening of the bolts permits the mounting plate **385** and the blade **300** to be locked at a desired angle relative to each other.

(117) Blade with Channels

(118) Referring to FIGS. 32 and 33, in certain embodiments the blade 300 has a plurality of channels 400. The channels 400 preferably are integral to the blade 300 in that the blade 300 defines walls 405 bounding the channels 400. When the blade 300 is attached to the base 100, the channels 400 (or at least some of the channels) on the blade 300 preferably extend generally away from the base 100.

(119) The channels 400 have a configuration designed to impact water flow along the blade. For example, the shape, size and orientation of the channels 400 may affect the shape of the wake. The channels 400 can be provided on either or both faces 305, 310 of the blade 300. The channels 400 can have different configurations on each face 305, 310 in order to provide different effects depending upon which face 305, 310 of the blade 300 faces the base 100.

(120) The channels 400 on a single blade 300 can all be uniform in size. Alternatively, the channels 400 on the blade 300 can be different in size. Where the channels 400 on the blade 300 are uniformly sized, the channels 400 can all be customized so they are smaller or larger as desired to affect performance.

(121) Additionally, the walls defining the channels 400 can optionally project outwardly from the blade 300 at a 90-degree angle, or they can form any angle with the blade 300 that is greater than zero. In certain embodiments, all the channels 400 on a given side of the blade are uniform in orientation and size. Alternatively, in some embodiments, the orientations of certain channels 400 on the blade 300 and/or the sizes of certain channels 400 are not uniform. Certain embodiments provide one side of the blade with channels of a given configuration, while channels on the other side of the blade have a different configuration.

(122) In FIGS. 32 and 33, all of the channels 400 are at least generally parallel to one other. Thus, the channels 400 can optionally be parallel to one another along an entire lengthwise extent of the blade 300. Although the channels 400 are shown as extending in straight lines across the blade 300, at least some (or all) of the channels 400 may instead be curved, e.g., one or more of them may have curved portions.

(123) The illustrated blade 300 further includes a mount region 410 configured to be carried against the base 100. The mount region 410 preferably is devoid of channels 400 so as not to interfere with mounting the blade 300 to the base 100. Depending on the particular embodiment, the mount region 410 can be an area on the blade 300 where an optional adapter component 325, mount structure 340, mounting plate 385, and/or base plate 390 are disposed.

(124) Deployable Lanyard Assembly

(125) In some embodiments, the invention provides a lanyard assembly 530. Referring to FIGS. 34-37, one example of the lanyard assembly 530 comprises a tether portion 535 and a float portion 540. The tether portion 535 comprises a tether line 555 configured for attaching to a cleat 560 on a boat 50. The tether line 555 can comprise a rope, bungee cord or the like. The float portion 540 comprises both a float 545 and a float line 550. The float line 550 can comprise a rope, bungee cord or the like and is configured for attaching to the base 100 of the apparatus 10. In some cases, the float line 550 can be attached to a lanyard eyelet 235 of the base 100.

(126) Preferably, the tether portion 535 and the float portion 540 are configured to be connected (and in FIG. 34, are connected) to each other by a friction joint. For example, the tether portion 535 can optionally comprise a male detent (e.g., a tongue) 565 and the float portion 540 can comprise a female detent 570. Alternatively, the male detent can be provided on the float portion while the female portion is provided on the tether position. In such cases, the male 565 and female 570 detents collectively define the friction joint. Various other friction joint structures can alternatively be used.

(127) The tether line 555 has first 575 and second 580 ends. The first end 575 of the tether line 555 is configured to be attached to a cleat 560 on a boat 50. The second end 580 of the tether line 555 can optionally have a tongue 585. The float line 550 also has first 590 and second 595 ends. The

first end **590** of the float line **550** is configured to be attached to the float **545**. The second end **595** of the float line **550** is configured to be attached to the base **100**. For example, in some cases the second end of the float line **550** is configured to be attached to a lanyard eyelet **235** of a wake shaping apparatus **10**. The float **545** has front **600** and rear **605** ends. In the embodiment illustrated, the rear end **605** of the float **545** has an opening **610**. The second end **595** of the illustrated float line **550** also has an opening **615**. The tongue **585** of the tether line **555** can be received in both openings **610**, **615** so as to provide a friction fit therebetween.

(128) The float **545** is buoyant such that it is configured to float on water. In a preferred embodiment, the float **645** is a foam material (e.g., a closed foam). However, the float **645** can comprise virtually any buoyant material, e.g., an inflatable plastic, so long as the float **545** is capable of providing buoyancy in water. As shown in FIGS. **36** and **37**, the float **545** has a bore (e.g., an axial bore) **630** extending between the front **600** and rear **605** ends thereof. Here, the tether line **555** extends through the bore **630** of the float **545**.

(129) Inflatable Wake Shaper Apparatus

(130) In certain embodiments, the invention provides a wake shaping apparatus **10** that is an inflatable apparatus **10b**. In such cases, the apparatus (at least when inflated) preferably can float on water. As used in the present disclosure, the term “inflatable apparatus” refers to an apparatus that can transition between a non-inflated state and an inflated state. The means for inflating the inflatable apparatus **10b** can be air or other gas, water or other liquid, or a combination of gas and liquid. In some cases, the apparatus **10b** is configured to have a varied internal pressure of gas or liquid (i.e., it can be inflated to different extents) to create multiple but intentional effects, such as different shapes and/or angles of the apparatus **10b**.

(131) Referring to FIGS. **38** and **39**, the inflatable apparatus **10b** has a plurality of walls **615**. In some embodiments, none of the walls **615** are rigid. In other embodiments, one or more of the walls **615** are rigid to help define a shape of the apparatus **10b**. For example, while being inflated, the apparatus **10b** can optionally expand until one of the rigid walls forms a desired angle with an adjacent non-rigid wall. The extent of expansion, and thus, the angle the rigid wall forms with the adjacent non-rigid wall, can optionally depend on a volume of gas and/or liquid introduced into the apparatus **10b**. Where the apparatus **10b** includes multiple rigid walls, the apparatus **10b** can be inflated to different extents such that the walls form different angles relative to each other. For instance, each rigid wall can form a unique, desired angle with an adjacent non-rigid wall. FIGS. **38** and **39** also show that the apparatus **10b** can in some cases be completely or substantially flattened when fully deflated, e.g., to permit compact storage of the apparatus **10b**.

(132) In some embodiments, the apparatus **10b** is empty and void of any internal structure. In other embodiments, the apparatus **10b** has inner chambers to provide rigidity, provide ballast and/or help define an inflated shape of the apparatus **10b**. Additionally, the apparatus **10b** can include internal structural elements that are independent of gas or liquid. Such internal structure could include drop stitch technology, in which tethers, such as strings or ropes, are attached to two or more walls **615** that define the shape of the apparatus **10b**. These tethers can optionally be flexible in a non-pressurized compressed environment. However, when the apparatus **10b** is inflated, the tethers may become taut and prevent the walls **615** from extending beyond an intentional length, thus confining the apparatus **10b** to a desired pressurized shape.

(133) As shown in FIG. **40**, the apparatus **10b** is used in combination with a boat **50** and can be attached to various locations on the boat **50**. For example, the apparatus **10b** can be attached to a single side **70**, **80** of the hull **40**. Alternatively, two inflatable apparatuses **10b** can be used such that one apparatus **10b** is attached to one side **70** of the hull **40** while another apparatus **10b** is attached to the other side **80** of the hull **40**. Each apparatus **10b** can be permanently or removably attached to the boat **50**. Preferably, the apparatus **10b** is attached to the boat **50** by at least one, and preferably two, suction cups **620**. This, however, is by no means required. It is also contemplated that the apparatus **10b** can include only a single suction cup **620**, or more than two suction cups **620**.

Alternatively, other removable and non-invasive forms of mechanical attachment, such as magnets, hook and loop fasteners or the like, can be provided for securing the apparatus **10b** to the hull **40** of a boat **50**.

(134) Where two inflatable apparatuses **10b** are used on a boat **50**, each apparatus **10b** can have an independent source (or a common source) of gas and/or liquid as the means for inflation. As shown in FIG. **40**, this source can include a controller **625** to regulate a volume of gas and/or liquid distributed into each apparatus **10b** by increasing or decreasing the pressure of the gas and/or liquid. These pressure changes can occur with the use of conventional valves in either an open or contained system. Such a system could have one or more gas or liquid pumps to operate the system to achieve desired levels. The system could also be an intake and dump-type siphon system, in which movement of the boat **50** is used as a means of intake to pressurize the system. Deflating or dumping pressure can be accomplished, for example, by pump or generator, natural aspiration allowing pressure inside the source container to equalize naturally, and/or with the assistance of motion of the boat **50** and force inflicted on the outside of the source containers.

(135) Such a system, as described above, would allow switching from operation of one inflatable apparatus **10b** on one side **70** of the hull **40** to operation of the other apparatus **10b** on the opposite side **80** of the hull **40** as needed. Thus, a selected one or more of the inflatable apparatuses **10b** can be inflated, or all of the inflatable apparatuses **10b** can remain in their uninflated states.

Advantageously, this system creates the desired effect of either virtually elongating one side **70**, **80** of the hull **40** or intentionally destroying wake development on one side of the boat **50**. Both effects can be used to generate mismatched wake conversion, and therefore, larger wake development for one side of the boat **50** relative to the other side of the boat **50**.

(136) In certain embodiments, the inflatable apparatus **10b** has at least one wall **615** that is concave. In the embodiment of FIGS. **41** and **42**, for example, a wall **615** (e.g., a top or leading wall) is concave so as to bound a channel **615C**. Configurations of this nature may provide special wake effects, e.g., by virtue of creating a somewhat channeled flow, which may have particular flow characteristics, such as laminar or substantially laminar flow through the channel. When the inflatable apparatus **10b** is mounted operatively to the hull of a boat, the wall **615** bounding the channel **615C** may face generally away from the hull and/or it may be oriented at an angle α (as measured relative to an axis normal to the hull surface to which the apparatus is attached and/or relative to a bottom wall **615** of the apparatus). The angle α convention here is best appreciated with reference to FIG. **39**.

(137) As shown in FIG. **39**, the inflatable apparatus **10b** can optionally include a leading wall **615** and a trailing wall **615** that are oriented at different angles α , θ . In some cases, angle α may be greater than 45 degrees, while angle θ is less than 45 degrees.

(138) The inflatable apparatus **10b** preferably has a wall **615** (e.g., a bottom wall) to which one or more (e.g., a plurality of) suction cups **620** are attached. In some cases, the one or more suction cups **620** are joined to a bottom wall **615** of the inflatable apparatus **10** by ultrasonic welding. In such cases, the suction cups are not intended to be separated from the bottom wall of the apparatus, but rather are integral/permanently attached to the bottom wall.

(139) In some embodiments, the inflatable apparatus **10b** includes a first wall **615** (e.g., a leading or front wall) that is rigid, a second wall **615** (e.g., a bottom wall) that is rigid, and one or more walls (e.g., side walls) **615** that are flexible. The one or more flexible walls may form (or be part of) a flexible, inflatable/collapsible enclosure, such as a PVC bag or another flexible bag or other bellows. The second wall **615** may be a bottom wall, and one or more suction cups may be attached to the bottom wall. Furthermore, if an apparatus of this nature has a channel **615C**, then the first wall **615** can optionally bound the channel.

(140) In embodiments where the wake shaping apparatus **10** is an inflatable apparatus **10b**, one or more of the walls **615** may be rigid. As just one example, such a rigid wall may comprise flexible top and bottom layers (e.g., comprising PVC or another polymer) between which there is

sandwiched a rigid sheet (e.g., comprising ABS or another rigid plastic).

(141) FIG. 43 shows another embodiment involving an inflatable apparatus **10b**. Here, the wake shaping apparatus **10** comprises a base **10bp** that can be removably secured to the hull **40** of a boat, and an inflatable apparatus **10b** that can be removably secured to the base **10bp**. The inflatable apparatus **10b** may be removably secured to the base **10bp** by virtue of, for example, the bottom wall **615** of the inflatable apparatus **10b** having two slots respectively configured to releasably receive and retain the two levers **195**. Additionally or alternatively, there can be another releasable mechanical attachment, such as hook and loop fasteners, clasps, magnets, or the like.

(142) As shown in FIG. 43, the base **10bp** preferably includes one or more (preferably a plurality of) suction cups **620**. In the embodiment illustrated, each suction cup **620** is operably coupled with a lever **195**, which can optionally be of the nature described elsewhere in this disclosure. Instead of levers, the suction cups can have other actuators, or may simply be configured to be pressed onto the hull without having actuators. Furthermore, instead of suction cups, the base can be removably attached to the hull through other non-invasive forms of mechanical attachment, such as hook and loop fasteners, clasps, magnets, or the like. Moreover, in some cases, the base **10bp** can have a different configuration so as to be attached permanently to the hull.

(143) In certain embodiments, the inflatable apparatus **10b** includes a wall **615** that is a leading (or “front”) wall, and thus faces generally away from the bottom of the apparatus and/or away from the hull **40**. If desired, this leading (or “front”) wall **615** can be a rigid wall having a flange **615F** projecting rearwardly of a rear wall (which may be a flexible, collapsible/expandable rear wall) of the inflatable apparatus. Reference is made to FIG. 43. While this figure involves a separable base **10bp**, it is to be appreciated that a rearwardly projecting flange **615F** can optionally be provided on an embodiment like the one shown in FIGS. 41 and 42, or on various other inflatable embodiments where there is no separable base.

(144) In some embodiments, the inflatable apparatus **10b** has one or more apertures **615H** formed so as to extend into the apparatus. Reference is made to FIGS. 44A-44C. Here, one or more apertures **615H** pass entirely from (and through) one wall **615** (optionally a sidewall) of the inflatable apparatus **10b** to (and through) another wall **615** (optionally an opposed sidewall) of the apparatus **10b**. In other cases, the apertures may be formed as blind bores. The aperture(s) **615H** preferably are exposed/open through one or more walls **615** of the inflatable apparatus **10b**. In some cases, each aperture **615H** opens through two walls **615** of the inflatable apparatus **10b**. When provided, the aperture(s) **615H** are configured to reduce buoyancy of the inflatable apparatus **10b**. This can make it easier to push under water, such as when mounting onto the hull **40** of a boat. Although not shown in FIGS. 44A-44C, there preferably are one or more suction cups or other mechanical attachments on the bottom of the inflatable apparatus in these embodiments.

(145) Thus, embodiments of the wake shaping apparatus are disclosed. One skilled in the art will appreciate that the present invention can be practiced with embodiments other than those disclosed. The disclosed embodiments are presented for purposes of illustration and not limitation, and the present invention is limited only by the claims that follow.

Claims

1. A wake shaping apparatus comprising a base and a blade, the base having first and second end regions and first and second side regions, the first and second side regions extending between the first and second end regions, the base having opposed top and bottom sides, the wake shaping apparatus configured such that an angular orientation of the blade is adjustable, relative to the base, by virtue of the blade being pivotable via a hinge, the wake shaping apparatus further comprising a plurality of suction cups on the bottom side of the base, the base having a maximum width, measured from the first side region of the base to the second side region of the base, and each suction cup having a diameter substantially equal to the maximum width of the base, each of the

suction cups having attached thereto a post, with each such post attached to a center of a respective one of the suction cups and extending outwardly therefrom along an axis, such that the suction cups have respective posts attached thereto and extending outwardly therefrom along respective axes, wherein each suction cup is coupled to a respective lever such that each lever is operable to activate and release suction for a respective one of the suction cups, each lever having a first position and a second position, such that when each lever is in the first position each suction cup is in a locked configuration, and when each lever is in the second position each suction cup is in an unlocked configuration.

2. The wake shaping apparatus of claim 1, wherein the blade projects away from the base.

3. The wake shaping apparatus of claim 1, wherein the blade has a width greater than the maximum width of the base.

4. The wake shaping apparatus of claim 1, wherein the base and the blade are attached to each other removably.

5. The wake shaping apparatus of claim 1, wherein the blade has a mount structure that projects outwardly from a face of the blade.

6. The wake shaping apparatus of claim 1, wherein the wake shaping apparatus is configured such that the angular orientation of the blade can be adjustable by pivoting the blade about a pivot axis that is parallel to the axes along which the posts extend.

7. The wake shaping apparatus of claim 1, wherein the suction cups remain in a constant plane when the angular configuration of the blade is adjusted.

8. The wake shaping apparatus of claim 1, wherein the plurality of suction cups comprises two suction cups, such that the wake shaping apparatus comprises two levers, a first of the two levers when in the second position having a free end pointing upwardly away from the base, and a second of the two levers when in the second position having a free end pointing upwardly away from the base, and wherein the first of the two levers when in the first position has its free end pointing in one direction, whereas the second of the two levers when in the first position has its free end pointing in another direction that is opposite said one direction.

9. The wake shaping apparatus of claim 1 wherein, rather than the base and the blade being attached to each other removably, the base is permanently affixed to the blade.

10. A wake shaping apparatus comprising a base and a blade, the base having first and second end regions and first and second side regions, the first and second side regions extending between the first and second end regions, the base having opposed top and bottom sides, the wake shaping apparatus configured such that an angular orientation of the blade is adjustable, wake shaping apparatus further comprising a plurality of suction cups on the bottom side of the base, each of the suction cups having attached thereto a post, with each such post attached to a center of a respective one of the suction cups and extending outwardly therefrom along an axis, such that the suction cups have respective posts attached thereto and extending outwardly therefrom along respective axes, wherein each suction cup is coupled to a respective lever such that each lever is operable to activate and release suction for a respective one of the suction cups, each lever having a first position and a second position, such that when each lever is in the first position each suction cup is in a locked configuration, and when each lever is in the second position each suction cup is in an unlocked configuration, wherein the base comprises a pivot connection between the first and second end regions, the pivot connection comprising a ball joint.

11. A wake shaping apparatus configured to be attached removably to a hull of a boat so as to disrupt water flow around the hull such that a wake created during operation of the boat is larger, differently position, and/or differently shaped than when the boat is operated without the wake shaping apparatus attached to the hull, the wake shaping apparatus comprising a blade and a base that has opposed first and second end regions and opposed top and bottom sides, the wake shaping apparatus further comprising a plurality of suction cups on the bottom side of the base, each of the suction cups being part of a suction cup assembly that further includes a post, a pin, and a lever,

whereby the wake shaping apparatus includes a plurality of posts, a plurality of pins, and a plurality of levers, such that each suction cup has attached thereto a respective one of the plurality of posts, each post has a hole therein, each lever has a hole therein, and each pin is received both in the hole in a respective one of the plurality of posts and in the hole in a respective one of the plurality of levers, wherein the base of the wake shaping apparatus has first and second side regions, the base has a maximum width, measured from the first side region of the base to the second side region of the base, and each suction cup has a diameter that is substantially equal to the maximum width of the base, the wake shaping apparatus further including a lanyard assembly comprising a float and a float line, the float line configured to be simultaneously attached to both the base and the float, such that the wake shaping apparatus is configured to float in a body of water, the wake shaping apparatus having a lanyard eyelet configured at the first or second side region of the base, wherein the lanyard eyelet is adjustable between a first position and a second position, such that when in the first position the lanyard eyelet is at the first side region of the base, and when in the second position the lanyard eyelet is at the second side region of the base.

12. The wake shaping apparatus of claim 11, wherein the float line is configured to be attached to an opening of the lanyard eyelet having a top open side exposed at the top side of the base and a bottom open side exposed at the bottom side of the base adjacent to an outer perimeter of one of the plurality of suction cups on the bottom side of the base.

13. The wake shaping apparatus of claim 11, wherein when the wake shaping apparatus is attached removably to a hull of a boat the opening of the lanyard eyelet to which the float line is configured to be attached is spaced apart from the hull by the plurality of suction cups such that a resulting clearance between the hull and the opening facilitates attachment of the float line to the opening of the lanyard eyelet.

14. The wake shaping apparatus of claim 11, wherein the float is a closed foam material.

15. The wake shaping apparatus of claim 11, wherein the float has a cylindrical configuration with a dome-shaped leading end.

16. The wake shaping apparatus of claim 11 wherein, rather than the base and the blade being attached to each other removably, the base is integral to the blade.

17. The wake shaping apparatus of claim 11, wherein the float line comprises a rope attached directly to both the float and the lanyard eyelet.
